



Food Quality Inspection System

Using Computer Vision & Machine Learning to automate fresh/rotten classification and quality scoring in real time.

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The Problem with Manual Inspection

Time-Consuming

Manual checking slows production lines and increases bottlenecks at scale.

Inconsistent

Human judgment varies by inspector, shift, and fatigue level — results are unreliable.

Error-Prone

Defects like early-stage rot or subtle discoloration are routinely missed by the naked eye.

Project Objective

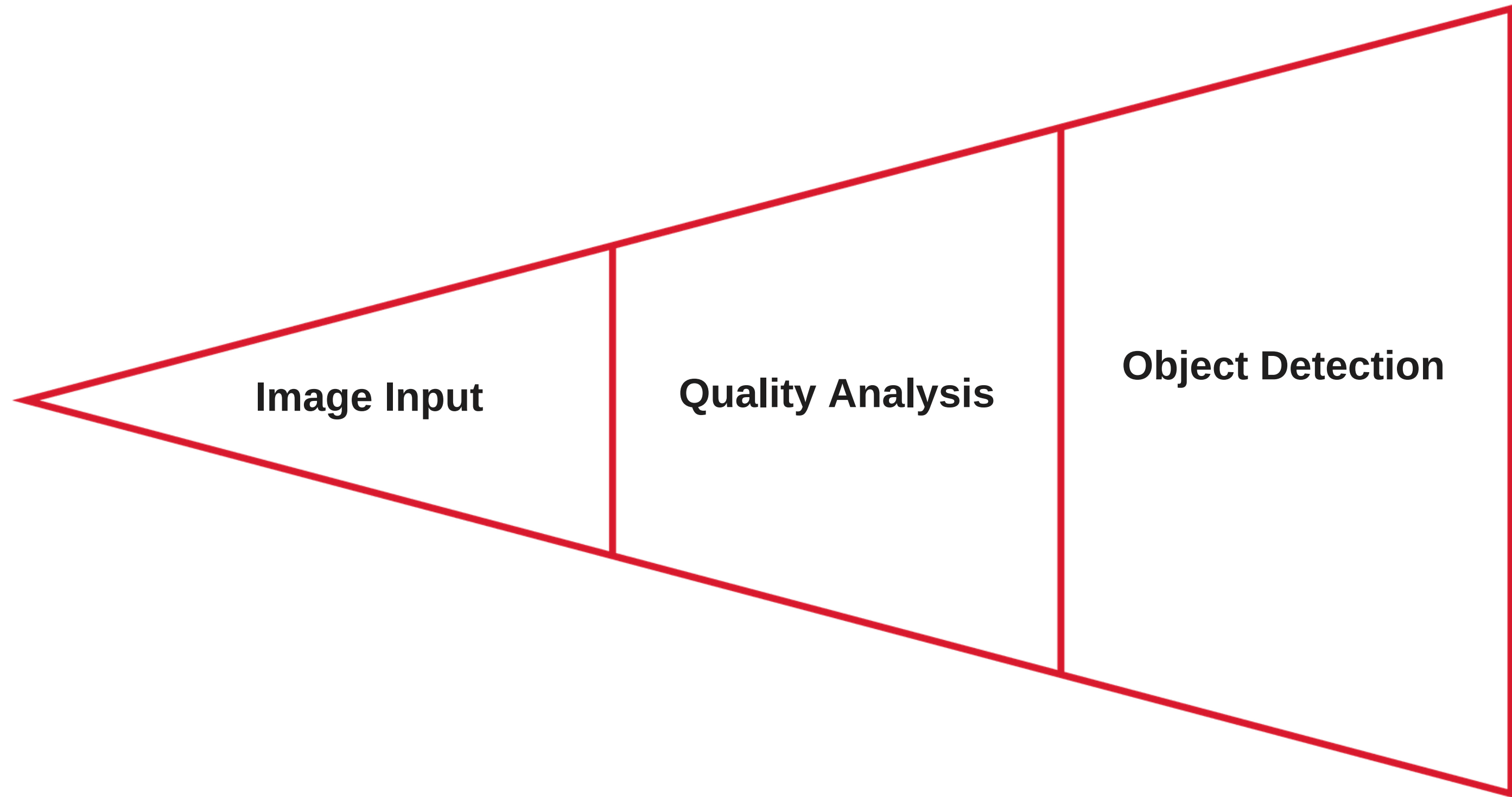
Core Goal

Build an automated system that classifies food as Fresh or Rotten and assigns a quality score — without any human intervention.

What the System Delivers

- Instant fresh/rotten classification
- Numerical quality score (0–100)
- Grade assignment: A, B, or C
- Clear Accept / Reject decision

How the System Works



Each image passes through detection, classification, and defect analysis before a final quality verdict is issued — all in seconds.

Technologies Used



Python

Core programming language powering the entire pipeline.



OpenCV

Image preprocessing, edge detection, and texture analysis.



PyTorch

Deep learning framework for training the classifier model.



Streamlit

Lightweight web UI for real-time image upload and result display.



YOLOv8

State-of-the-art object detection to localize the food region.

Key Features

Real-Time Processing

Images are analyzed instantly — no batch delays or waiting queues.

Automated Quality Scoring

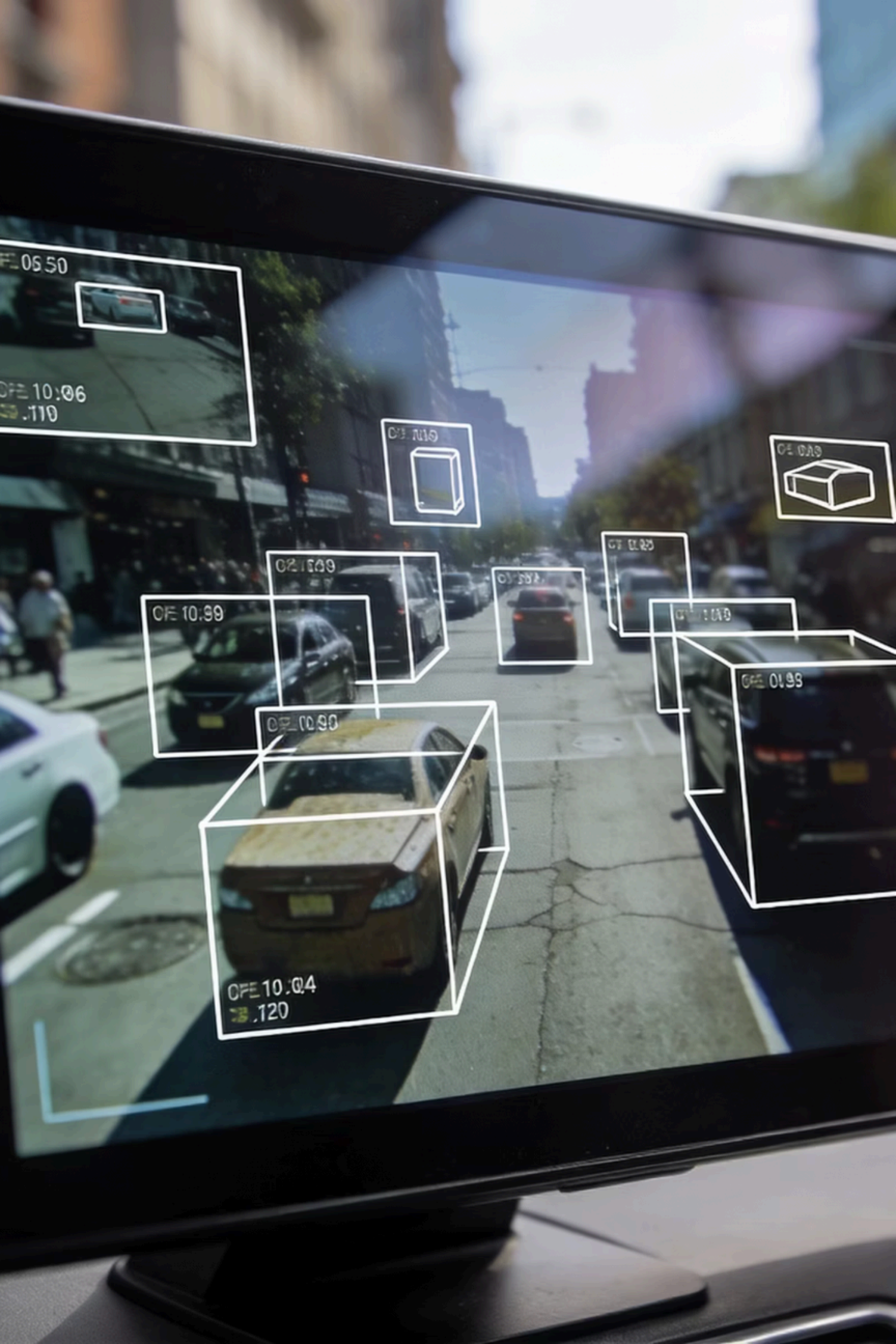
Every item receives a 0–100 score and a letter grade without human input.

Defect Detection

Identifies dark spots, texture irregularities, and edge anomalies that indicate rot.

User-Friendly Interface

Simple drag-and-drop upload via Streamlit — no technical expertise required to operate.



Live Demo Walkthrough

01

Upload Image

User selects a fruit/food photo via the Streamlit interface.

02

YOLO Detection

System draws a bounding box around the food region and extracts the ROI.

03

Analysis & Score

Classifier + defect engine produce a quality score, grade, and Accept/Reject verdict.

Challenges & Solutions

Challenges Faced

- *Limited dataset* — labeled food quality images are scarce and hard to source.
- *Lighting variation* — inconsistent lighting changes perceived color and texture dramatically.
- *Visual similarity* — early-stage rotten items closely resemble fresh ones.

How We Addressed Them

- ✓ Combined heuristic-based scoring with ML predictions to improve robustness even with limited training data.

This hybrid approach ensures the system remains reliable when the model encounters edge cases or ambiguous inputs.

Future Scope



Larger Dataset

Train on diverse, large-scale datasets to boost model generalization.



Industry Deployment

Integrate with real production lines for automated inline quality control.



Enhanced Accuracy

Leverage advanced deep learning architectures to reduce misclassification.



Broader Coverage

Extend beyond fruits to vegetables, grains, and packaged food products.



Conclusion

This system replaces slow, inconsistent manual inspection with a fast, AI-driven pipeline — delivering reliable quality decisions at scale.

Automated No human intervention needed for classification or scoring.	Fast Real-time results from upload to verdict in seconds.
Consistent Same reliable standard applied to every single item inspected.	